Kraft's inequality

A detailed proof of the subtree construction

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September 6, 2026

The converse of Kraft's inequality requires more than enough space for the desired subtrees: their leaves must also be aligned with the branches of a single binary tree. The construction below makes this alignment explicit. The decisive fact is that, after the lengths have been sorted, every earlier block size is an integer multiple of the next one.

Theorem 1 (Kraft's inequality)**.** *Let $K \geq 1$ and let $\ell_1, \dots, \ell_K$ be nonnegative integers. There exist binary words $c_1, \dots, c_K$ with $|c_k| = \ell_k$ such that, for distinct j, k, c_j is not a prefix of c_k, if and only if*

$$\sum_{k=1}^{K} 2^{-\ell_k} \leq 1. \tag{1}$$

Here, a word is a prefix of itself, so the condition also rules out assigning the same word to two different symbols. Length zero means the empty word, denoted by ε. If some $\ell_k = 0$, (1) forces $K = 1$, and the empty word gives the required code. For coding a message over that one-symbol alphabet, its length must be supplied separately, as in the book.

Which consecutive leaves form a subtree?

Fix $m \geq 0$. Consider the complete binary tree whose vertices are all words of length at most m; the children of w are $w\mathtt{0}$ and $w\mathtt{1}$. Its depth-m leaves are the words in $\{0,1\}^m$. Number these leaves from left to right by $0, 1, \dots, 2^m - 1$, using their values as binary integers.

For a word $w = w_1 \cdots w_\ell$ of length $\ell \leq m$, put

$$q(w) = \sum_{r=1}^{\ell} w_r 2^{\ell - r}, \qquad q(\varepsilon) = 0,$$

and let $\mathcal{D}_m(w)$ be the set of indices of its depth-m descendants. Leading zeroes are part of the word even though they do not change $q(w)$.

Lemma 1 (Aligned blocks)**.** *A word w of length ℓ has exactly the following descendant leaves:*

$$\mathcal{D}_m(w) = \big\{ q(w)2^{m-\ell}, \dots, (q(w)+1)2^{m-\ell} - 1 \big\}. \tag{2}$$

Consequently, a consecutive block of $b = 2^{m-\ell}$ leaves, starting at index S and contained in $\{0, \dots, 2^m - 1\}$, is the descendant-leaf set of a depth-ℓ vertex if and only if b divides S.

Proof. Every descendant word is obtained by appending $m - \ell$ bits to w. If those appended bits represent the integer r, its leaf index is $q(w)2^{m-\ell} + r$. As the appended bits vary, r takes each value from 0 to $2^{m-\ell} - 1$ exactly once, proving (2).

Thus a descendant block starts at a multiple of b. Conversely, if $S = qb$ and the block fits, then $0 \leq q < 2^\ell$. Write q in binary using exactly ℓ bits, adding leading zeroes if necessary. Equation (2) shows that this word has precisely the prescribed descendant block. For $\ell = 0$, the word is ε and its block consists of all 2^m leaves. □

Proof of Kraft's inequality

Necessity. Suppose the words are prefix-free, and set $m = \max_k \ell_k$. The sets $\mathcal{D}_m(c_k)$ are pairwise disjoint. Indeed, if a depth-m word extended both c_j and c_k, its two prefixes would be comparable: the shorter would be a prefix of the longer. This contradicts prefix-freeness when $j \neq k$. The aligned-block lemma therefore gives

$$\sum_{k=1}^{K} 2^{m-\ell_k} = \sum_{k=1}^{K} |\mathcal{D}_m(c_k)| \leq 2^m.$$

Division by 2^m gives (1).

Sufficiency. Assume (1). Relabel the symbols so that

$$\ell_1 \leq \ell_2 \leq \cdots \leq \ell_K, \qquad m = \ell_K.$$

We will assign a vertex of the *same complete depth-m tree* to each symbol. No rearrangement of the tree's branches is involved.

1. **Specify the blocks.** For each k, define the block size and cumulative endpoints by

$$b_k = 2^{m-\ell_k}, \qquad S_0 = 0, \qquad S_k = \sum_{j=1}^{k} b_j, \qquad B_k = \{S_{k-1}, \ldots, S_k - 1\}. \tag{3}$$

   These are nonempty, pairwise disjoint sets of consecutive indices. Moreover,

$$S_K = 2^m \sum_{k=1}^{K} 2^{-\ell_k} \leq 2^m,$$

   so every B_k fits inside the leaf set of the ambient tree.

2. **Verify alignment.** For $j < k$, the ordering gives $\ell_j \leq \ell_k$, hence

$$\frac{b_j}{b_k} = 2^{\ell_k - \ell_j} \in \mathbb{N}.$$

   Thus *each earlier block size is divisible by b_k*, and so is their sum:

$$\frac{S_{k-1}}{b_k} = \sum_{j<k} 2^{\ell_k - \ell_j} \in \mathbb{Z}_{\geq 0}. \tag{4}$$

   For $k = 1$, the sum is empty and equals zero. This is the precise reason that consecutive packing respects the branching structure of the tree.

3. **Turn each block into a word.** Set $q_k = S_{k-1}/b_k$. It is an integer by (4), and

$$0 \leq q_k, \qquad (q_k + 1)b_k = S_k \leq 2^m, \qquad \text{so} \qquad 0 \leq q_k \leq 2^{\ell_k} - 1.$$

   There is therefore a unique word c_k of exactly ℓ_k bits representing q_k, with leading zeroes included. By Lemma 1,

$$\mathcal{D}_m(c_k) = B_k. \tag{5}$$

   This constructs actual vertices of the original complete tree, rather than merely blocks with the correct cardinalities.

4. **Verify the prefix property.** If c_j were a prefix of c_k for distinct indices, every depth-m extension of c_k would also extend c_j. Consequently,

$$\varnothing \neq B_k = \mathcal{D}_m(c_k) \subseteq \mathcal{D}_m(c_j) = B_j,$$

contradicting the disjointness of B_j and B_k. The words are therefore prefix-free. Undoing the initial relabeling assigns each original symbol a word of its prescribed length.

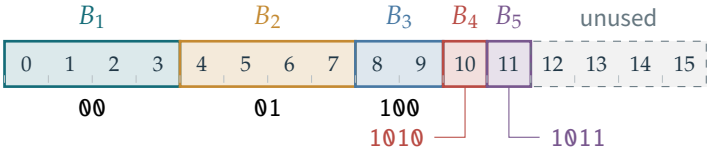
This proves the converse and completes the proof of the theorem.

Recovering the coding tree. To obtain the book's representation $c = \mathrm{Leaves}(T)$, take T to be the union of the paths from the root to $c_1, \ldots, c_K$. A codeword has no strict descendant in T, since otherwise it would be a prefix of another codeword. Conversely, a vertex of T that is not a codeword lies strictly before some c_k on its path and therefore has a child in T. The leaves of T are exactly the selected codewords. This pruned coding tree may have vertices with only one child; it need not be complete.

A concrete example

Take $(\ell_1, \ldots, \ell_5) = (2, 2, 3, 4, 4)$. Then $m = 4$ and $\sum_k 2^{-\ell_k} = 3/4$. The construction is entirely explicit:

k	ℓ_k	b_k	S_{k-1}	$q_k = S_{k-1}/b_k$	c_k
1	2	4	0	0	`00`
2	2	4	4	1	`01`
3	3	2	8	4	`100`
4	4	1	10	10	`1010`
5	4	1	11	11	`1011`



Each colored block contains the depth-4 extensions of the word written below it. In particular, $B_3 = \{8, 9\}$ corresponds to the vertex `100`, since its extensions are `1000` and `1001`. The four unused leaves cause no difficulty: Kraft's inequality does not require that every leaf of the ambient tree be assigned to a codeword.

Why the sorting matters. For lengths $(3, 1)$, the Kraft sum is $5/8$, so a code exists. However, at depth $m = 3$, packing blocks in that order gives $\{0\}$ followed by $\{1, 2, 3, 4\}$. The latter block is not the descendant set of a depth-1 vertex: the only such blocks are $\{0, 1, 2, 3\}$ and $\{4, 5, 6, 7\}$. Sorting the lengths as $(1, 3)$ gives blocks $\{0, 1, 2, 3\}$ and $\{4\}$, corresponding to `0` and `100`. They can then be reassigned to the original symbols in the required order.

An alternative inductive proof of the converse

The same argument can be phrased without consecutive blocks. After sorting the lengths, suppose that prefix-free words $c_1, \dots, c_{k-1}$ of the prescribed lengths have been chosen. At depth ℓ_k, the extensions of c_j occupy exactly $2^{\ell_k - \ell_j}$ vertices. These forbidden sets are disjoint, because the previously chosen words are prefix-free. Hence the number of available vertices is

$$A_k = 2^{\ell_k} - \sum_{j<k} 2^{\ell_k - \ell_j}.$$

The partial Kraft inequality gives

$$\sum_{j<k} 2^{-\ell_j} + 2^{-\ell_k} \le 1, \qquad \text{and therefore} \qquad A_k \ge 1.$$

Choose an available word at depth ℓ_k. It cannot extend any previous codeword, by availability. It cannot be a prefix of one either: all previous lengths are at most ℓ_k, and equality of words is also excluded by availability. Thus the enlarged family is prefix-free, and induction constructs all K codewords.

This version separates the two ingredients particularly clearly: sorted lengths prevent a newly chosen word from sitting above an earlier one, and Kraft's inequality guarantees a free vertex at every step.